
Case Study 15

Design and Implementation of a Medical Question-and-Answer Assistant Using Generative AI and Large Language Models

An Applied Solution Design Combining BERT-Based Understanding, GPT-Style Generation, Retrieval-Augmented Grounding, LoRA/PEFT Fine-Tuning and RLHF Safety Alignment

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This report analyses the problem of patient-facing medical question answering, justifies the selection of a hybrid BERT + GPT + RAG + LoRA/PEFT + RLHF architecture, and presents an implementation roadmap, expected outcomes, and limitations.

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1. Executive Summary

Healthcare systems worldwide face a growing gap between patient demand for accessible medical information and the availability of qualified clinicians to answer routine, non-emergency health queries. This case study presents an end-to-end design for a Medical Question and Answer (Q&A) Assistant built on Generative AI and Large Language Model (LLM) technology. The proposed system combines a pretrained transformer-based generative model, retrieval-augmented generation (RAG) over curated medical knowledge bases, parameter-efficient fine-tuning (LoRA/PEFT) on domain-specific corpora, and Reinforcement Learning from Human Feedback (RLHF) to align responses with medical safety and accuracy standards. The report analyses the problem, justifies the model and architecture choices, describes an implementation roadmap, and critically evaluates the expected outcomes and limitations of deploying such a system in a real-world clinical or consumer-health setting.

2. Introduction and Background

Generative AI, powered by transformer-based Large Language Models such as the GPT family and encoder models such as BERT, has demonstrated strong capability in understanding and generating natural language across domains. In healthcare, patients increasingly turn to search engines and chatbots to understand symptoms, medications, and general wellness questions before or instead of consulting a physician. Traditional rule-based chatbots and FAQ systems are brittle, cannot handle free-form natural language queries, and struggle to generalise to the near-infinite variety of ways a patient may phrase a health concern.

A Medical Q&A Assistant built on modern generative language models can understand nuanced, conversational queries, retrieve relevant clinical knowledge, and generate coherent, contextually appropriate answers. However, because incorrect or hallucinated medical information can cause real harm, such a system must be engineered with far greater emphasis on factual grounding, safety alignment, and transparency than a general-purpose chatbot.

3. Problem Statement

A healthcare-technology organisation wants to deploy a virtual assistant that can answer patient questions about symptoms, medications, general disease information, and preventive care in natural language, twenty-four hours a day. The assistant must:

- Understand free-text medical questions phrased in everyday language, including spelling variations and colloquial symptom descriptions.

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- Provide accurate, evidence-based, and up-to-date medical information sourced from trusted references.
 - Clearly communicate its limitations and refer users to a licensed medical professional for diagnosis, prescriptions, or emergencies.
 - Avoid hallucinating facts, dosages, or drug interactions, since incorrect answers could cause patient harm.
 - Preserve patient privacy and comply with healthcare data-protection regulations such as HIPAA and India's Digital Personal Data Protection Act.
 - Scale to handle a large volume of concurrent users at low latency and reasonable cost.

The challenge, therefore, is to design a Generative AI solution that balances fluent, empathetic natural-language interaction with the strict factual reliability and safety requirements unique to the medical domain.

4. Requirement Analysis

Before selecting a model or architecture, the functional and non-functional requirements of the system were analysed as follows.

4.1 Functional Requirements

- Accept natural-language medical questions via text (and optionally voice) input.
- Classify query intent (symptom check, medication information, general health education, appointment triage, emergency detection).
- Retrieve relevant, verified medical content from a trusted knowledge base.
- Generate a fluent, empathetic, and medically appropriate response with citations where possible.
- Escalate or flag queries that indicate a medical emergency or require a licensed professional.

4.2 Non-Functional Requirements

- High factual accuracy and low hallucination rate.
- Low response latency (target under three seconds for typical queries).
- Data privacy, encryption, and regulatory compliance.
- Auditability — every answer should be traceable to a source document.

- Cost-efficient fine-tuning and inference, given limited computational budget compared to training a model from scratch.

5. Technology Landscape: Comparing Candidate Approaches

Several classes of language model architecture were evaluated for suitability against the requirements above.

Approach	Strength	Weakness	Suitability
BERT (encoder-only)	Excellent at classification, intent detection, entity extraction	Cannot generate free-form conversational answers	Used for intent/triage classifier, not the answer generator
GPT-style decoder LLM	Fluent generation, strong reasoning, conversational tone	Prone to hallucination if used without grounding	Core generative engine, used with retrieval grounding
Full fine-tuning of an LLM	Deep domain adaptation	Extremely costly, risk of catastrophic forgetting	Not cost-effective for this project
LoRA / PEFT fine-tuning	Cheap, fast, modular adaptation to medical corpus and tone	Slightly lower ceiling than full fine-tuning	Selected for domain adaptation
RLHF alignment	Improves safety, tone, refusal behaviour on risky queries	Requires curated human preference data and reward modelling	Selected for final safety alignment layer
Retrieval-Augmented Generation (RAG)	Grounds answers in verified sources, reduces hallucination, easy to update	Answer quality depends on retrieval quality	Selected as the core grounding mechanism

6. Model Selection and Justification

Based on the comparison above, a hybrid architecture is proposed rather than relying on a single model:

6.1 BERT-based Intent and Entity Classifier

A fine-tuned BERT (or a lightweight distilled variant such as DistilBERT / BioBERT / ClinicalBERT) is used as a front-end classifier. BioBERT and ClinicalBERT are pretrained on biomedical literature and clinical notes respectively, giving them a strong prior understanding of

medical terminology. This classifier performs intent detection (symptom query, medication query, emergency query, general education) and named-entity recognition to extract symptoms, drug names, and body parts mentioned in the query. This step is fast, cheap, and highly accurate for classification-style tasks, and it allows the system to route emergency-sounding queries to a safety-override path before any generative response is produced.

6.2 GPT-style Decoder LLM as the Generative Engine

A GPT-style autoregressive transformer (for example, an open-weight instruction-tuned model in the same family as GPT, LLaMA, or Mistral) is used to compose the final natural-language answer. Decoder-only models are selected over encoder-only models such as BERT for this stage because the task requires open-ended text generation, multi-turn dialogue coherence, and the ability to summarise retrieved passages into a patient-friendly explanation — capabilities that encoder-only architectures do not natively provide.

6.3 Retrieval-Augmented Generation (RAG)

Rather than relying purely on the model's parametric memory, the system retrieves the most relevant passages from a curated, regularly updated medical knowledge base (for example, MedlinePlus, WHO fact sheets, national drug formularies, and institution-approved clinical guidelines) using dense vector similarity search. The retrieved passages are inserted into the LLM's prompt context, so that the generated answer is grounded in verified text and can cite its source. This substantially reduces hallucination compared to using the LLM alone.

6.4 LoRA / PEFT for Domain Adaptation

Low-Rank Adaptation (LoRA), a Parameter-Efficient Fine-Tuning (PEFT) technique, is used to adapt the base GPT-style model to the medical domain without updating all of its billions of parameters. LoRA freezes the pretrained weights and injects small trainable low-rank matrices into the attention layers, dramatically reducing the number of trainable parameters (often by more than ninety-nine percent) and the GPU memory required. This makes it feasible to fine-tune on a moderate medical Q&A dataset using limited institutional compute, while preserving the base model's general language ability and reducing the risk of catastrophic forgetting.

6.5 RLHF for Safety and Tone Alignment

After LoRA-based domain adaptation, Reinforcement Learning from Human Feedback is applied as a final alignment stage. Clinicians and safety reviewers rank candidate answers to sample questions; a reward model is trained on these rankings; and the generative model is fine-tuned (for example, using Proximal Policy Optimisation) to prefer answers that are accurate, appropriately cautious, empathetic in tone, and that correctly refuse or redirect unsafe requests such as dosage

calculations for self-medication or requests for a definitive diagnosis. RLHF is particularly valuable in this domain because it directly encodes human clinical judgement about what a safe answer looks like, which cannot be fully captured by supervised fine-tuning alone.

7. Proposed System Architecture

The end-to-end pipeline for a single user query proceeds through the following stages:

- 1. Input handling — the patient's question is received via text or speech-to-text and normalised (spelling correction, language detection).
- 2. Safety pre-filter — a lightweight classifier scans for emergency keywords (for example, chest pain, suicidal ideation, severe bleeding) and immediately routes such cases to an emergency-guidance response and helpline information, bypassing generation.
- 3. Intent and entity extraction — the BERT-based classifier identifies query type and extracts medical entities.
- 4. Retrieval — a dense vector search over the medical knowledge base returns the top-k relevant passages.
- 5. Prompt construction — the retrieved passages, extracted entities, and conversation history are assembled into a structured prompt for the generative model.
- 6. Generation — the LoRA-fine-tuned, RLHF-aligned GPT-style model produces a draft answer grounded in the retrieved context.
- 7. Post-processing and guardrails — an automated checker verifies the answer does not contain unsupported claims, dosage instructions, or a definitive diagnosis, and appends a disclaimer and source citations.
- 8. Response delivery — the final answer is returned to the user, with an option to escalate to a human clinician or telehealth service.

This layered design ensures that the flexible, fluent generative model is always constrained by retrieval grounding and rule-based safety checks, rather than operating as an unconstrained free-text generator.

8. Implementation Approach

8.1 Data Collection and Preparation

Training and evaluation data would be drawn from publicly available, licensed medical Q&A datasets (for example, MedQuAD, PubMedQA, and MedMCQA), supplemented with de-identified, institution-approved patient query logs where available. All data undergoes de-identification, medical review, and quality filtering before use. The knowledge base for retrieval is built from trusted structured sources such as clinical guidelines, drug formularies, and public-health fact sheets, chunked into passages and embedded using a sentence-embedding model for vector search.

8.2 Fine-Tuning Pipeline

The base GPT-style model is fine-tuned in two stages. First, supervised fine-tuning with LoRA adapters is performed on curated medical question-answer pairs, teaching the model the tone, structure, and terminology expected of a medical assistant. Second, an RLHF stage is applied: multiple candidate answers are sampled for each prompt, ranked by clinician reviewers, a reward model is trained on these preferences, and the policy model is updated using reinforcement learning so that its outputs increasingly match clinically preferred answers.

8.3 Retrieval Infrastructure

A vector database (for example, FAISS, Pinecone, or a managed equivalent) stores embeddings of the knowledge-base passages. At inference time, the user query embedding is compared against this index to retrieve the most semantically relevant passages, which are injected into the LLM prompt following a retrieval-augmented generation pattern.

8.4 Evaluation Before Deployment

Prior to release, the system undergoes automated benchmarking against held-out medical Q&A test sets, as well as structured clinician review of a sample of generated answers for factual correctness, safety, and appropriateness of tone. A red-teaming phase specifically probes the system with adversarial and edge-case queries (attempts to obtain dosage instructions, self-diagnosis pressure, and emotionally sensitive scenarios) to verify guardrails function correctly.

8.5 Deployment and Monitoring

The system is deployed behind an API layer with logging, rate limiting, and authentication. Continuous monitoring tracks hallucination reports, user feedback ratings, and escalation rates. The knowledge base is refreshed on a scheduled basis as clinical guidelines are updated, and

periodic re-alignment (further RLHF rounds) is performed as new failure modes are discovered in production.

8.6 Evaluation Metrics

- Factual accuracy rate — proportion of generated answers judged clinically correct by reviewer panels against source guidelines.
- Hallucination rate — proportion of statements in an answer that are not supported by any retrieved passage.
- Retrieval precision/recall — whether the passages fetched for a query actually contain the information needed to answer it.
- Safety-trigger accuracy — correct identification and routing of emergency or high-risk queries during red-team testing.
- Response latency — end-to-end time from query submission to answer delivery.
- User satisfaction and trust scores — collected through in-app feedback ratings and periodic surveys.

9. Expected Outcomes

- Patients receive fast, twenty-four-hour access to reliable, plain-language answers to common, non-urgent health questions.
- Reduced burden on clinical staff and call centres for routine informational queries, freeing time for higher-acuity care.
- Improved patient understanding of medications, symptoms, and preventive care, supported by cited sources.
- A measurable reduction in hallucinated or unsupported claims compared to an ungrounded LLM, due to retrieval augmentation and guardrails.
- Safer handling of high-risk queries, with consistent redirection to emergency services or licensed professionals where appropriate.
- A modular system that can be updated with new medical knowledge without full retraining, since the knowledge base is decoupled from model weights.

10. Limitations and Risks

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- Residual hallucination risk: even with retrieval grounding, generative models can occasionally produce plausible-sounding but incorrect statements, particularly for rare conditions or ambiguous queries.
 - No substitute for diagnosis: the system cannot examine a patient physically or interpret diagnostic imaging or laboratory results, and must not be used for definitive diagnosis or treatment decisions.
 - Bias in training data: medical literature and historical patient records can under-represent certain populations, potentially leading to less accurate or less relevant answers for those groups.
 - Regulatory and liability complexity: deploying an AI system that gives health information raises questions about medical liability, regulatory classification (as a medical device in some jurisdictions), and the need for ongoing clinical oversight.
 - Privacy risk: handling sensitive patient queries requires strict data governance; any breach could expose protected health information.
 - Dependence on knowledge-base quality: the accuracy of retrieval-augmented answers is only as good as the curated knowledge base; outdated or poorly curated sources will degrade answer quality.
 - Language and literacy barriers: the system may perform less well for underrepresented languages, dialects, or low-literacy phrasing unless specifically trained and tested for these cases.
 - Computational cost of RLHF: collecting high-quality clinician preference rankings at scale is time-consuming and expensive, which can slow iteration on safety alignment.

10.1 Illustrative Interaction Walkthrough

To make the pipeline concrete, consider a patient who types: "I have had a mild headache and slight fever for two days, what should I do?" The safety pre-filter scans the query and finds no emergency keywords, so it proceeds. The BERT-based classifier tags the intent as a symptom-query and extracts the entities headache and fever with a two-day duration. The retrieval module searches the knowledge base and returns passages on common causes of headache with low-grade fever and guidance on when to seek in-person care. These passages are inserted into the prompt along with the extracted entities, and the LoRA-fine-tuned, RLHF-aligned generative model composes a response that explains likely benign causes, suggests supportive self-care such as hydration and rest, lists specific warning signs that should prompt an urgent visit, and includes a disclaimer recommending consultation with a physician if symptoms persist beyond a few days or worsen. The post-processing guardrail confirms the answer does not recommend a specific drug

dosage, and the response is delivered with citations to the source passages. This walkthrough illustrates how each architectural layer contributes a distinct safeguard: intent classification for routing, retrieval for grounding, fine-tuning for domain tone, RLHF for cautious framing, and rule-based post-checks for a final safety net.

11. Ethical Considerations

Deploying a Medical Q&A Assistant carries ethical obligations beyond technical accuracy. The system must be transparent about being an AI and not a licensed clinician, must avoid creating false confidence in its answers, and must be designed with informed consent regarding data use. Regular audits for demographic bias in answer quality, clear escalation paths to human care, and mechanisms for patients to report incorrect answers are essential components of responsible deployment, not optional add-ons.

12. Conclusion

A safe and effective Medical Q&A Assistant cannot rely on a single generative model used in isolation. The solution proposed here combines a BERT-based classifier for intent and entity recognition, a GPT-style decoder model as the fluent generative engine, retrieval-augmented generation for factual grounding, LoRA-based parameter-efficient fine-tuning for cost-effective domain adaptation, and RLHF for safety and tone alignment. This layered, guardrail-driven architecture balances the flexibility and fluency of modern LLMs with the strict reliability, transparency, and safety requirements that the medical domain demands, while remaining computationally and financially feasible to build and maintain.

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